# Template

Studentnames and studentnumbers here

2026-06-17

## Set-up your environment

```r
require(tidyverse)
```

```
## Loading required package: tidyverse

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```r
require(yaml)
```

```
## Loading required package: yaml
```

```r
library(dplyr)
library(systemfonts)
```

## Title Page

Changes in nitrogen emissions in the Netherlands

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4.3

Fatima Amankour

# Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

## 1.1 Describe the Social Problem

Include the following:

- Why is this relevant?
- Nitrogen emissions represent a critical public health crisis in the Netherlands. Elevated atmospheric concentrations lead to the production of particulate manner and ozone, which are closely associated with serious health problems such as worsening asthma, long-term cardiovascular problems, and diminished lung function.
- This report measures the correlation between public health concerns and regional nitrogen levels over almost 20 years, from 2005 to 2023. By analyzing these temporal and spatial patterns, we want to pinpoint pollution hotspots and offer a data-driven framework for focused reduction initiatives, like industrial restrictions or localized agriculture changes, to successfully lessen the burden on public health.

# Part 2 - Data Sourcing

## 2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```r
PolutionInGeneral<- read.csv("data/Polution in general.csv")
BA_BronnenVanStikstof <- read.csv("data/Bronnen van stikstof.csv")
AA_BronnenVanFijnstof <- read.csv("data/Bronnen van Fijnstof.csv")
JaargemiddeldeFijnstof <- read.csv("data/Jaargemiddelde PM10 per Locatie 2015-2025.csv", sep = ";")
Meetlocaties <- read.csv("data/luchtmeetnet_meetlocaties.csv")


saveRDS(PolutionInGeneral, "data/PolutionInGeneral.rds")
PolutionInGeneral <- readRDS("data/PolutionInGeneral.rds")
saveRDS(BA_BronnenVanStikstof, "data/BA_BronnenVanStikstof.rds")
BA_BronnenVanStikstof <- readRDS("data/BA_BronnenVanStikstof.rds")
saveRDS(AA_BronnenVanFijnstof, "data/AA_BronnenVanFijnstof.rds")
AA_BronnenVanFijnstof <- readRDS("data/AA_BronnenVanFijnstof.rds")
saveRDS(JaargemiddeldeFijnstof, "data/JaargemiddeldeFijnstof.rds")
JaargemiddeldeFijnstof <- readRDS("data/JaargemiddeldeFijnstof.rds")
saveRDS(Meetlocaties, "data/Meetlocaties.rds")
Meetlocaties <- readRDS("data/Meetlocaties.rds")
```

## 2.2 Provide a short summary of the dataset(s)

```r
#head(PolutionInGeneral)
```

This dataset contains 11 variables, each representing a different year. For each year, data is provided on the levels of nitrogen, NMVOCs (non-methane volatile organic compounds), sulfur oxides, and PM2.5 particulate matter observed in the air. By analyzing this data, we can assess the severity of air pollution and identify trends in air quality over time.

```r
#head(JaargemiddeldeFijnstof)
```

These datasets show particulate matter (fine dust) levels measured at various monitoring stations across the Netherlands over the years 2005 till 2025. This provides a clear insight into the periods and regions where particulate matter causes the greatest impact and pollution.

```r
#head(BA_BronnenVanStikstof)
```

This dataset shows the sources of nitrogen measured over a time period from 2005 till 2024

```r
#head(AA_BronnenVanFijnstof)
```

This dataset shows the sources of fine dust measured over a time period from 2005 till 2024

```r
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

### 2.3 Describe the type of variables included

Most of the datasets include a duration of time most often 2005 till 2024, and a variable that may be a specific measuring station or a source of polution.

# Part 3 - Quantifying

### 3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```r
#Laatste rij was .cbs, die hebben we verwijderd
AA_BronnenVanFijnstof<- AA_BronnenVanFijnstof [-nrow(AA_BronnenVanFijnstof),]

AA_BronnenVanFijnstof <- AA_BronnenVanFijnstof %>%
  mutate(across(where(is.character),~ as.numeric(gsub(",",".", .x))))

#Nieuwe variabele gemaakt
AA_BronnenVanFijnstof$Totaal <- AA_BronnenVanFijnstof$Landbouw..bosbouw.en.visserij..mln.kg.+ AA_Bronne
```

```
BA_BronnenVanStikstof<- BA_BronnenVanStikstof [-nrow(BA_BronnenVanStikstof),]

BA_BronnenVanStikstof <- BA_BronnenVanStikstof %>%
  mutate(across(where(is.character),~ as.numeric(gsub(",",".", .x))))


BA_BronnenVanStikstof$Totaal <- BA_BronnenVanStikstof$Landbouw..bosbouw.en.visserij..mln.kg.+ BA_Bronnen
```

**Cleans the Meetlocaties dataset by splitting all data into separate columns and removing the first four rows.**

```
# We overschrijven 'Meetlocaties' direct zodat de oude versie verdwijnt
Meetlocaties <- Meetlocaties %>%
  # 1. Haal de rijen met een '#' direct weg
  filter(!str_starts(.[[1]], "#")) %>%

  # 2. Splits direct op de puntkomma's naar losse kolommen
  separate_wider_delim(
    cols = 1,
    delim = ";",
    names = c("meetlocatie_id", "bron_id", "meetlocatie_naam",
              "meetlocatie_plaatsnaam", "breedtegraad", "lengtegraad",
              "hoogte", "meetlocatie_begindatumtijd", "meetlocatie_einddatumtijd")
  )
```

**Puts Meetlocaties & JaargemiddeldeFijnstof together the data gets a clear location**

```
library(tidyverse)

# 1. Zorg dat de koppelnaam in JaargemiddeldeFijnstof exact matcht
names(JaargemiddeldeFijnstof)[1] <- "meetlocatie_id"

# 2. Voer de koppeling uit en pak flexibel alle kolommen mee
Fijnstof_Met_Locaties <- JaargemiddeldeFijnstof %>%
  # Koppel de schone Meetlocaties tabel eraan vast
  left_join(Meetlocaties, by = "meetlocatie_id") %>%

  # De nieuwe indeling: we zetten de handige kolommen vooraan,
  # en met 'everything()' komen ALLE overige kolommen (de jaren) er automatisch achteraan!
  select(
    Plaatsnaam = meetlocatie_plaatsnaam,
    Breedtegraad = breedtegraad,
    Lengtegraad = lengtegraad,
    everything(),          # Neemt automatisch alle overige kolommen/jaren mee
    -meetlocatie_id,       # Haalt de oude NL-code netjes weg
    -bron_id,              # Ruimt direct overbodige kolommen uit Meetlocaties op
    -meetlocatie_naam,
    -hoogte,
    -meetlocatie_begindatumtijd,
```

```
    -meetlocatie_einddatumtijd
  )
```

**Makes Fijnstof_Met_Locaties numeric**

```
Fijnstof_Met_Locaties <- Fijnstof_Met_Locaties %>%
  mutate(
    # 1. Coördinaten omzetten naar numeriek
    Breedtegraad = as.numeric(Breedtegraad),
    Lengtegraad = as.numeric(Lengtegraad)
  ) %>%
  mutate(
    # 2. Alle jaarkolommen (die beginnen met X en daarna 4 cijfers) in één keer opschonen.
    # Dit haalt de sterretjes, spaties en streepjes weg en maakt er getallen van.
    across(
      .cols = matches("^X\\d{4}$"),
      .fns = ~ as.numeric(str_replace_all(.x, "([*]|\\s|-)", ""))
    )
  )
```

2 datasets naast elkaar gezet, en dat per jaar vergeleken, dataset fijnstof gaat tot 2024 en dataset stikstof gaat tot 2023, daar moest ik nog een aanpassing in maken.

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

## 3.2 Generate necessary variables

**Creates total variables**

```
# De bestaande kolom 'Totaal' wordt gebruikt.
# De totalen van fijnstof en stikstof worden gecombineerd op basis van het jaar.
# Omdat stikstof tot en met 2023 loopt, loopt de gecombineerde tabel ook tot en met 2023.

AB_Fijnstof_Totaal <- data.frame(
  Jaar = AA_BronnenVanFijnstof[, 1],
  Totaal_fijnstof = AA_BronnenVanFijnstof$Totaal
)

BB_Stikstof_Totaal <- data.frame(
  Jaar = BA_BronnenVanStikstof[, 1],
  Totaal_stikstof = BA_BronnenVanStikstof$Totaal
)

totale_data <- merge(
  AB_Fijnstof_Totaal,
  BB_Stikstof_Totaal,
  by = "Jaar"
)
```

**DIstancing data from the randstad from the Non-randstad**

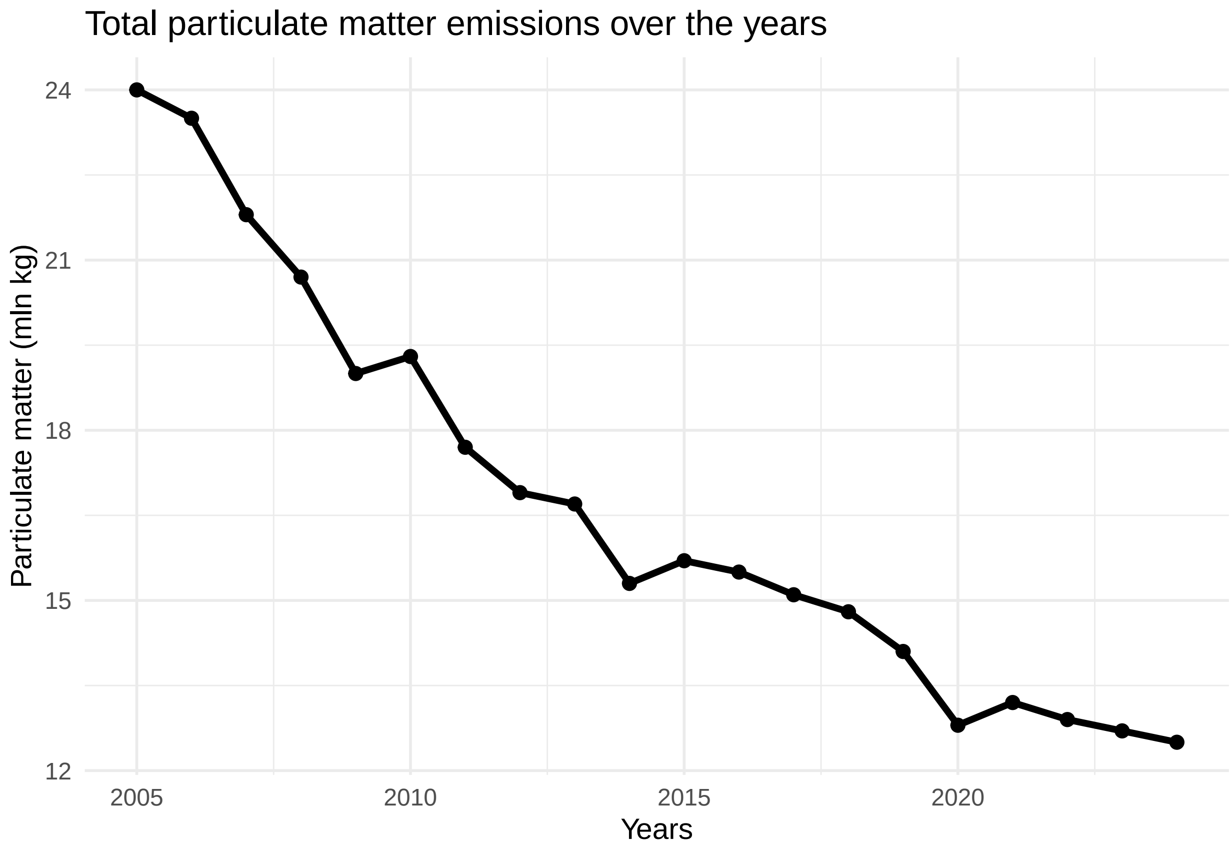
```r
# Maak eerst de basisregio-dataset aan (als je dat nog niet had gedaan)
randstad_plaatsen <- c("Amsterdam", "Rotterdam", "Den Haag", "Utrecht",
                       "Haarlem", "Zaanstad", "Leiden", "Dordrecht",
                       "Almere", "Amersfoort", "Delft", "Schiedam")

Fijnstof_Regio <- Fijnstof_Met_Locaties %>%
  mutate(Regio = if_else(Plaatsnaam %in% randstad_plaatsen, "Randstad", "Other_regions"))

# --- HIER SPLITSEN WE ZE IN TWEE LOSSE DATASETS ---
Fijnstof_Randstad      <- Fijnstof_Regio %>% filter(Regio == "Randstad")
Fijnstof_Other_regions <- Fijnstof_Regio %>% filter(Regio == "Other_regions")
```

## 3.3 Create line graphs

**Simple graph about the general emission of particulate matter (fijnstof)**

This graph shows the evolution of total particulate matter emissions over the years from 2005 to 2025, where you can see that the total particulate matter emissions have almost halved over the years.

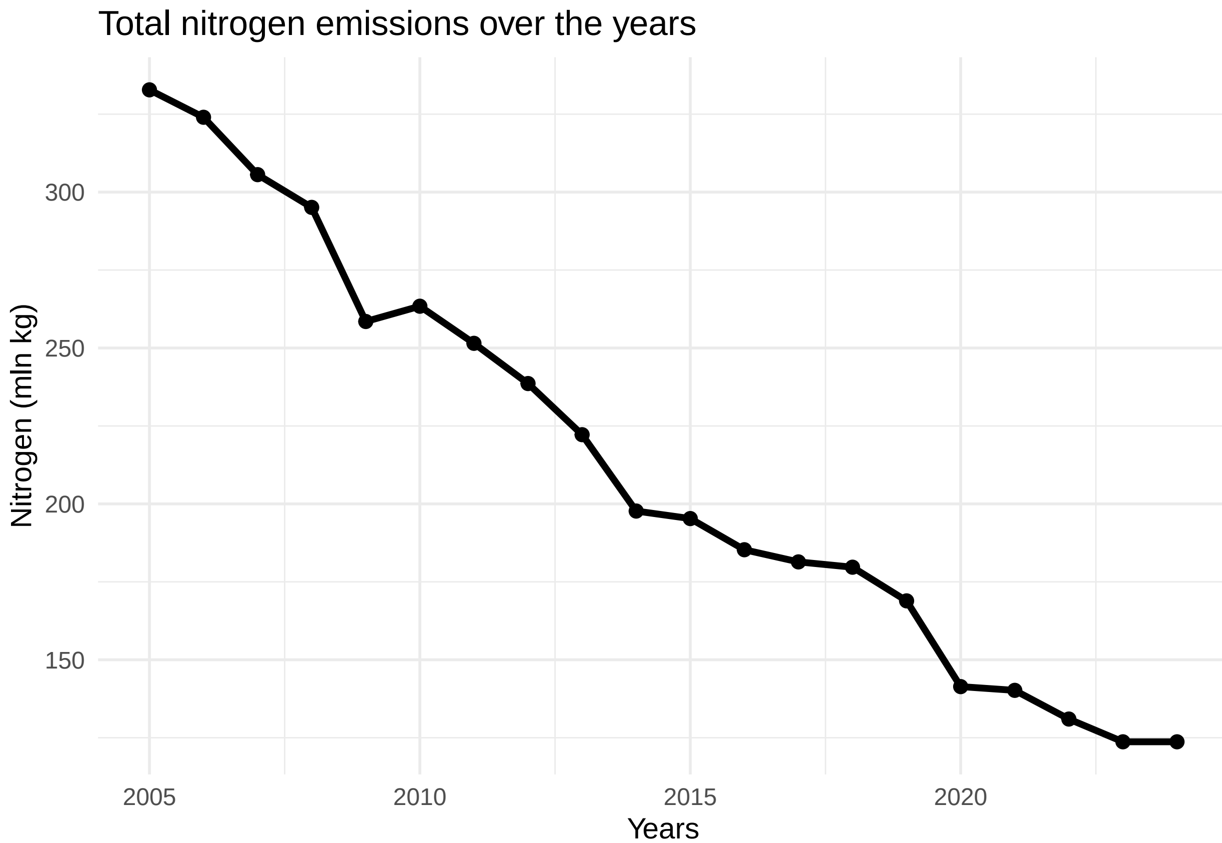
```r
library(ggplot2)

ggplot(AA_BronnenVanFijnstof, aes(x = X., y = Totaal)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Total particulate matter emissions over the years",
    x = "Years",
    y = "Particulate matter (mln kg)"
  ) +
  theme_minimal()
```




**Simple graph about the general emission of nitrogen**

This graph shows the evolution of the emission of nitrogen over the years from 2005 to 2025, where you can see that the total particulate matter emissions have more than halved over the years.

```r
library(ggplot2)

ggplot(BA_BronnenVanStikstof, aes(x = X., y = Totaal)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Total nitrogen emissions over the years",
    x = "Years",
    y = "Nitrogen (mln kg)"
  ) +
  theme_minimal()
```

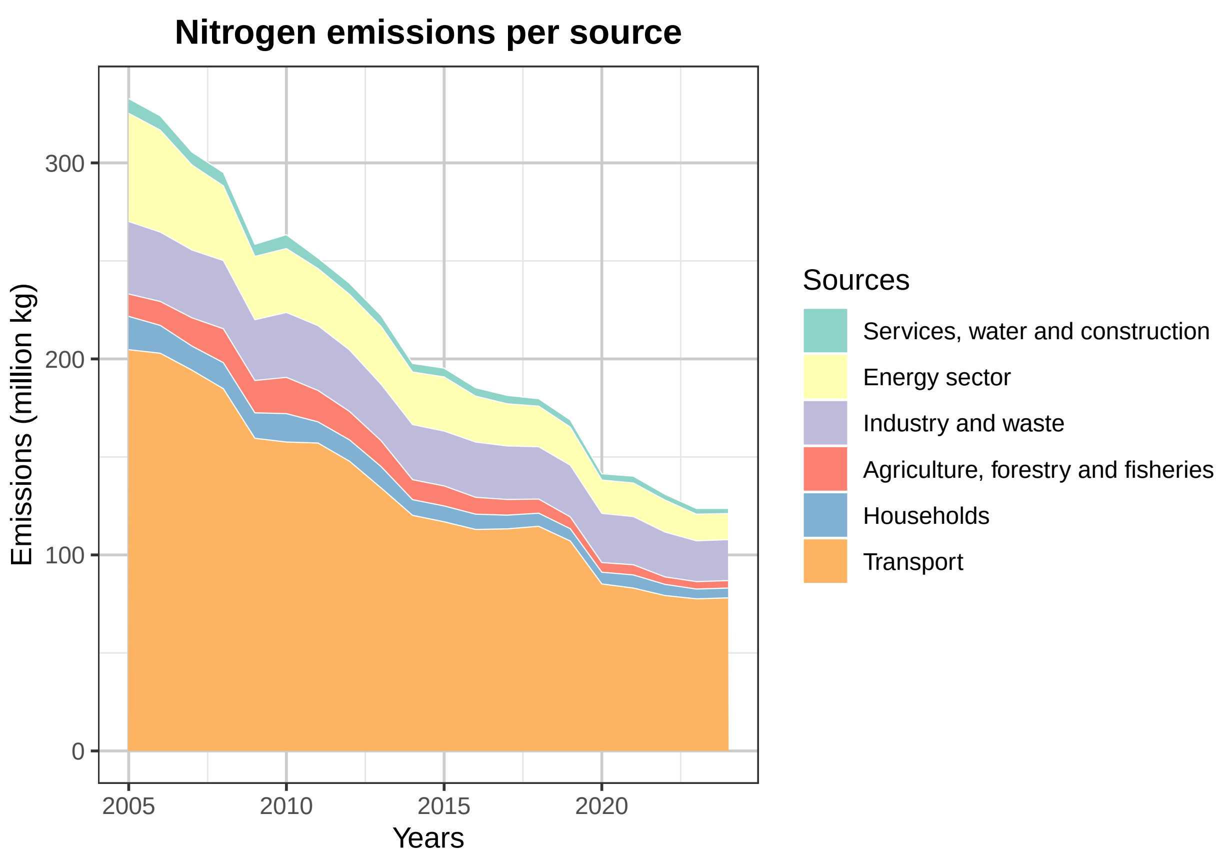



**Detailed and Multifaceted Map of Nitrogen Emission Origins**

A more detailed graph displays the different sources of nitrogen emissions using distinct colors, making it easier to distinguish between the various emission sources.

```
# 1. Laad de benodigde bibliotheken
library(tidyverse)

# 2. Bereid de dataset voor: hernoem de eerste kolom naar "X."
names(BA_BronnenVanStikstof)[1] <- "X."

# 3. Zet de tabel om naar het lange formaat dat jouw ggplot verwacht
BA_BronnenVanStikstof_long <- BA_BronnenVanStikstof %>%
  pivot_longer(
    cols = -X.,
    names_to = "Bron",
    values_to = "Uitstoot"
  ) %>%
  # BELANGRIJK: Filter de "Totaal"-kolom eruit zodat deze de grafiek niet vertekent
  filter(Bron != "Totaal")


ggplot(BA_BronnenVanStikstof_long,
       aes(x = X.,                                     # <--- Werkt nu met de jaren uit je CSV
           y = Uitstoot,
```

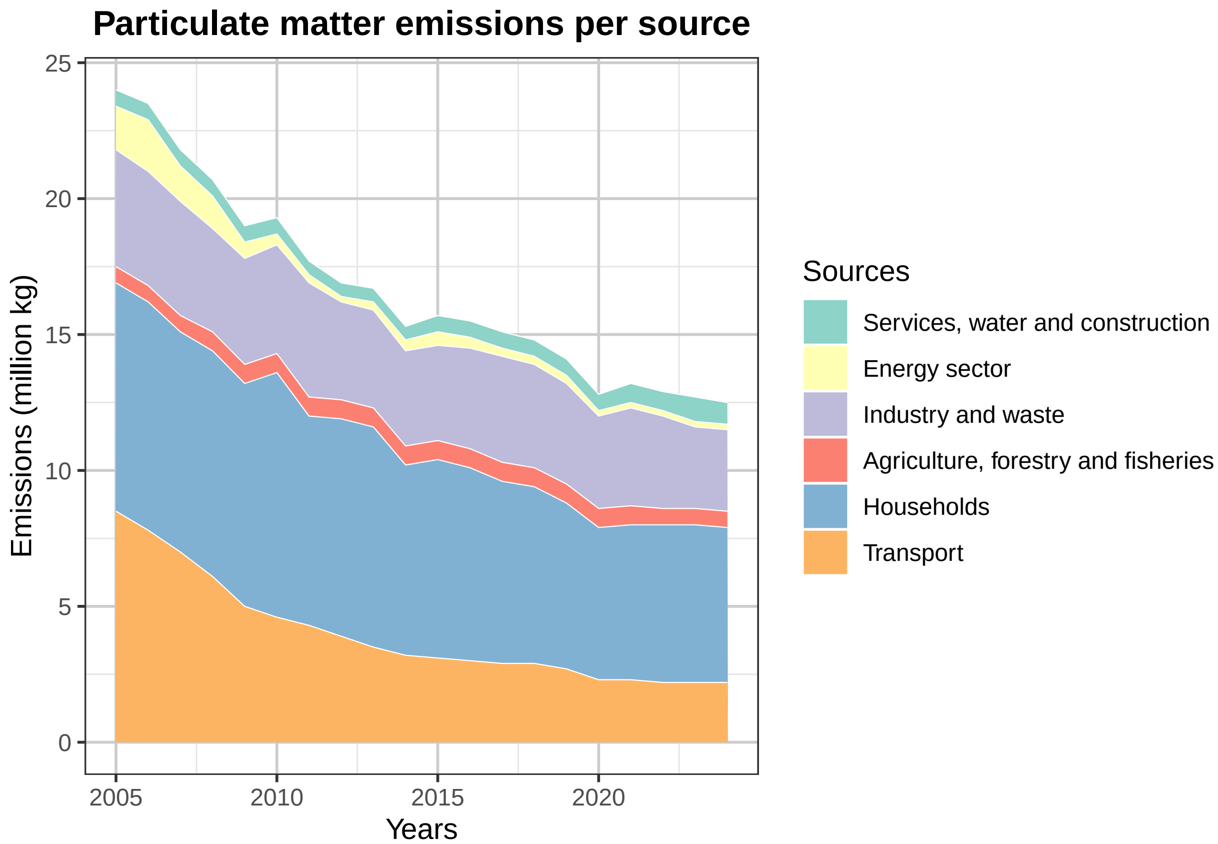
```
           fill = Bron)) +
  geom_area(color = "white", linewidth = 0.2) +       # <--- Creëert de witte scheidingslijntjes
  scale_fill_brewer(
  palette = "Set3",
  labels = c(
    "Services, water and construction",
    "Energy sector",
    "Industry and waste",
    "Agriculture, forestry and fisheries",
    "Households",
    "Transport"
  )
) +
  labs(                                               # <--- Namen op de assen en titels
    title = "Nitrogen emissions per source",
    x = "Years",
   y = "Emissions (million kg)",
    fill = "Sources"
  ) +
  theme_bw() +                                        # <--- Grijze rasterlijnen inschakelen
  theme(
    panel.grid.major = element_line(color = "grey80"),
    panel.grid.minor = element_line(color = "grey90"),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```



**Detailed and Multifaceted Map of Particulate matter Origins**

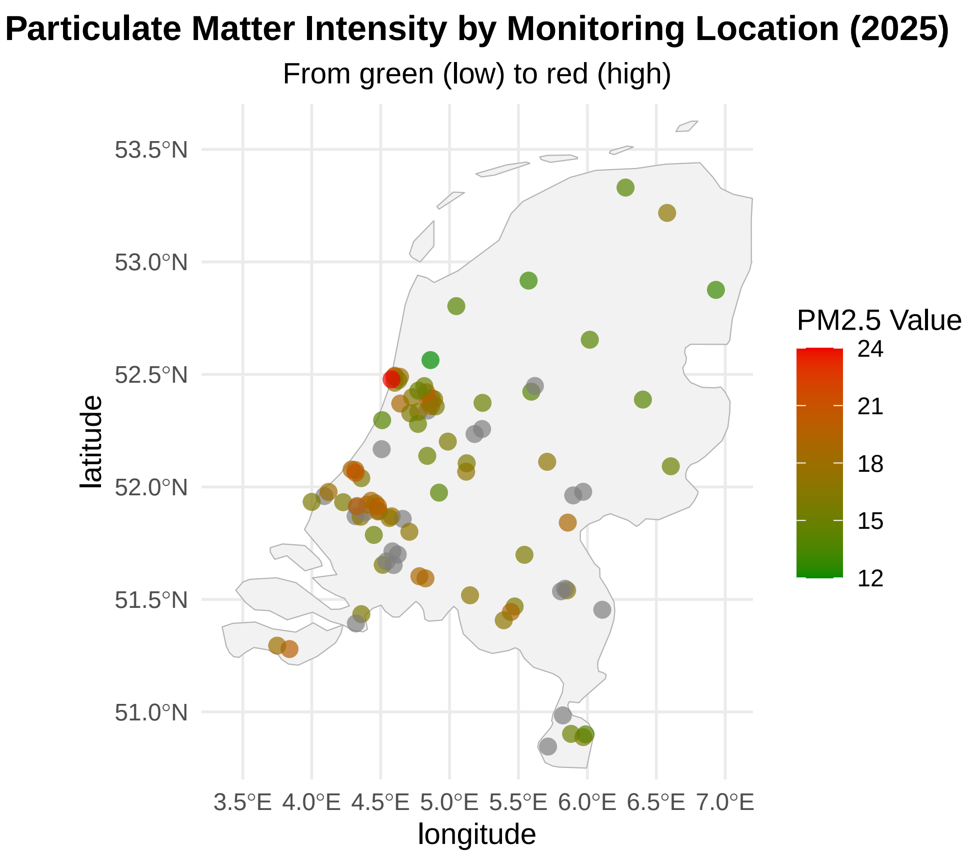
A more detailed graph displays the different sources of fine particulate matter emissions using distinct colors, making it easier to distinguish between the various emission sources.

```
# 1. Laad de benodigde bibliotheken
library(tidyverse)

# 2. Bereid de dataset voor: hernoem de eerste kolom naar "X."
names(AA_BronnenVanFijnstof)[1] <- "X."

# 3. Zet de tabel om naar het lange formaat dat jouw ggplot verwacht
AA_BronnenVanFijnstof_long <- AA_BronnenVanFijnstof %>%
  pivot_longer(
    cols = -X.,
    names_to = "Bron",
    values_to = "Uitstoot"
  ) %>%
  # BELANGRIJK: Filter de "Totaal"-kolom eruit zodat deze de grafiek niet vertekent
  filter(Bron != "Totaal")


ggplot(AA_BronnenVanFijnstof_long,
       aes(x = X.,                                    # <--- Werkt nu met de jaren uit je CSV
           y = Uitstoot,
           fill = Bron)) +
  geom_area(color = "white", linewidth = 0.2) +       # <--- Creëert de witte scheidingslijntjes
  scale_fill_brewer(
  palette = "Set3",
  labels = c(
    "Services, water and construction",
    "Energy sector",
    "Industry and waste",
    "Agriculture, forestry and fisheries",
    "Households",
    "Transport"
  )
) +
  labs(                                               # <--- Namen op de assen en titels
    title = "Particulate matter emissions per source",
    x = "Years",
    y = "Emissions (million kg)",
    fill = "Sources"
  ) +
  theme_bw() +                                        # <--- Grijze rasterlijnen inschakelen
  theme(
    panel.grid.major = element_line(color = "grey80"),
    panel.grid.minor = element_line(color = "grey90"),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```




### 3.4 Creating a heatmap

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```




This map displays the locations of air quality monitoring stations across the Netherlands and shows the measured concentrations of fine particulate matter in the air for the year 2025. A conventional heatmap, in which all municipalities are represented by different colors, was not feasible because data were available for only a limited number of municipalities. In addition, larger municipalities often contained multiple monitoring stations, which sometimes recorded substantially different measurements due to local environmental conditions and station-specific locations. As a result, displaying the individual monitoring stations provides a more accurate representation of the available data.

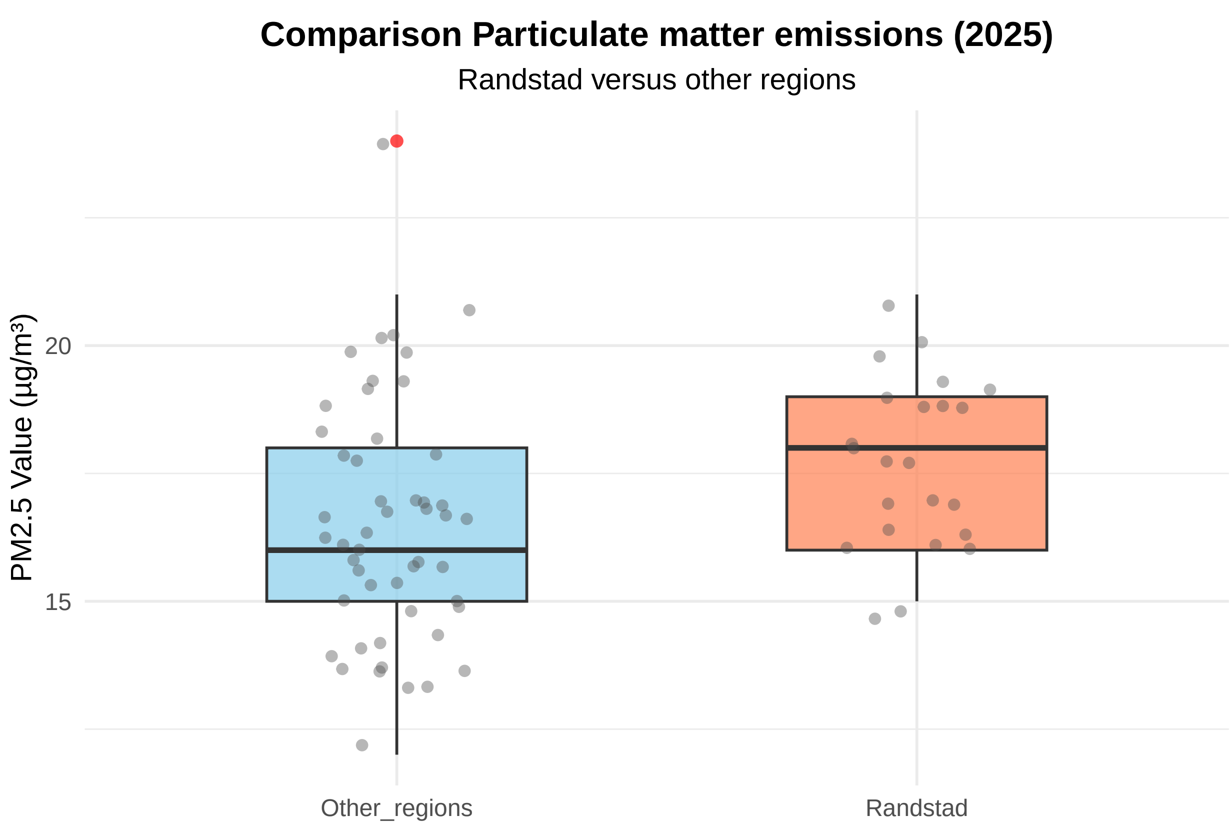
## 3.5 Visualize sub-population variation

#### A boxplot that combines the randstad and the rest

```r
library(tidyverse)

# We gebruiken de gecombineerde dataset 'Fijnstof_Regio'
ggplot(data = Fijnstof_Regio, aes(x = Regio, y = as.numeric(`X2025`), fill = Regio)) +

# 1. Teken de boxplots naast elkaar (voeg na.rm = TRUE toe)
geom_boxplot(width = 0.5, alpha = 0.7,
             outlier.color = "red", outlier.shape = 16, outlier.size = 2,
             na.rm = TRUE) +

# 2. Voeg de individuele meetstations als stipjes toe (voeg na.rm = TRUE toe)
geom_jitter(width = 0.15, alpha = 0.4, color = "grey30",
```

```r
           na.rm = TRUE) +

  # 3. Geef beide groepen een eigen herkenbare kleur
  scale_fill_manual(values = c("Randstad" = "coral", "Other_regions" = "skyblue")) +

  scale_x_discrete(labels = c("Niet-Randstad" = "Other regions")) +
  # 4. Titels en labels aanpassen
  labs(
    title = "Comparison Particulate matter emissions (2025)",
    subtitle = "Randstad versus other regions",
    y = "PM2.5 Value (µg/m³)",
    x = "" # De X-as laat nu netjes de categorienamen zien
  ) +

  # 5. De lay-out strak maken (de labels op de X-as laten we nu wél staan!)
  theme_minimal() +
  theme(
    legend.position = "none", # Legenda is overbodig omdat de X-as het al vertelt
    plot.title = element_text(face = "bold", hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5)
  ) +
  scale_x_discrete(labels = c("Niet-Randstad" = "Other regions", "Randstad" = "Randstad"))
```

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



By comparing the PM 2.5 particulate matter levels between the randstad boxplot and the other regions boxplot, the boxplot comparison clearly shows regional differences in Dutch air quality. This graphic directly relates to the project's main social issue, which is examining the effects of transportation, industrial infrastructure, and densely populated areas on localized pollution and public health.

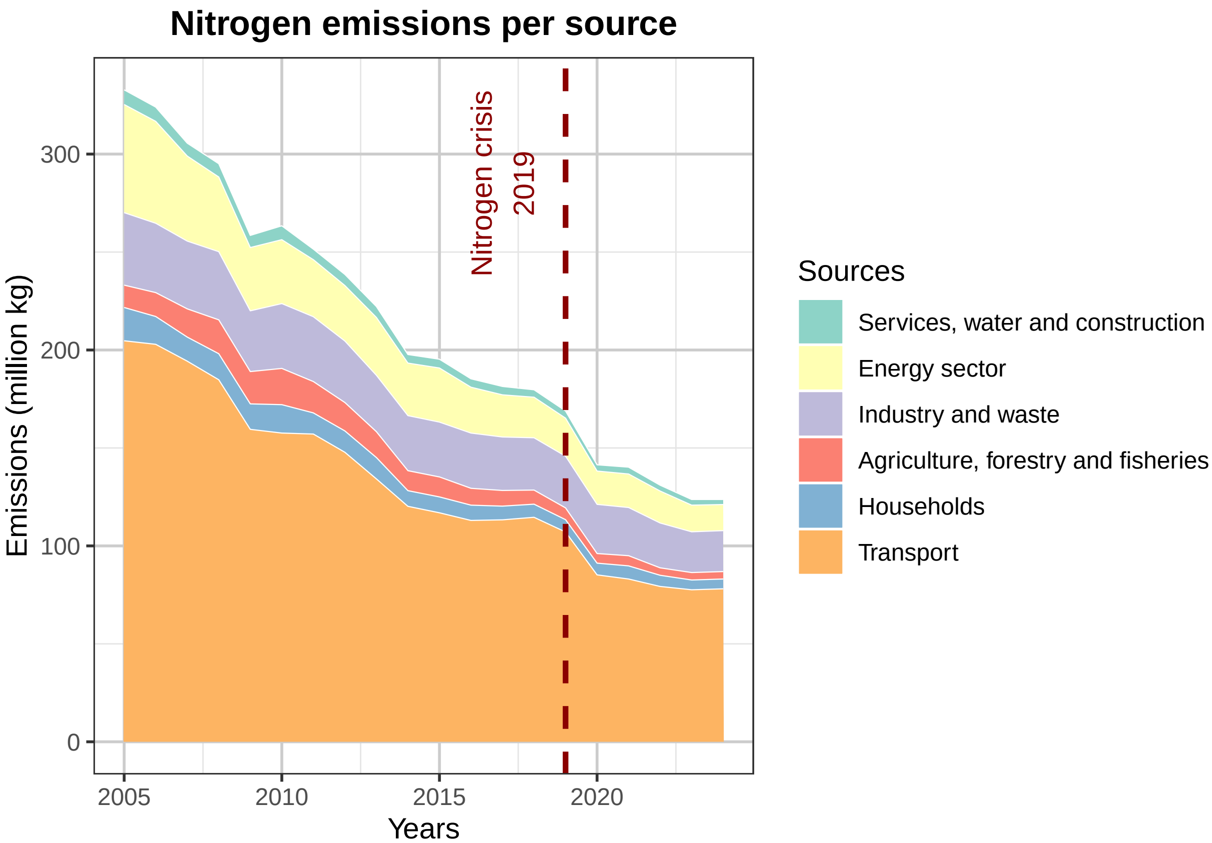
These plots demonstrate that randstad areas have a greater, more concentrated load of particle matter, even though national data indicates that industries like agriculture dominate nitrogen emissions across rural regions. The randstad boxplot's larger median PM 2.5 value highlights the unequal distribution of environmental effects and presents air pollution as a pressing public health and urban development issue.

### 3.6 Event analysis

Analyze the relationship between two variables.

```r
# 1. Laad de benodigde bibliotheken
library(tidyverse)

# 2. Bereid de dataset voor: hernoem de eerste kolom naar "X."
names(BA_BronnenVanStikstof)[1] <- "X."

# (Optioneel: Zorg dat je jaartallen numeriek zijn, mocht de lijn niet werken)
# BA_BronnenVanStikstof$X. <- as.numeric(BA_BronnenVanStikstof$X.)

# 3. Zet de tabel om naar het lange formaat dat jouw ggplot verwacht
BA_BronnenVanStikstof_long <- BA_BronnenVanStikstof %>%
  pivot_longer(
    cols = -X.,
    names_to = "Bron",
    values_to = "Uitstoot"
  ) %>%
  # BELANGRIJK: Filter de "Totaal"-kolom eruit zodat deze de grafiek niet vertekent
  filter(Bron != "Totaal")


# 4. Maak de plot
ggplot(BA_BronnenVanStikstof_long,
       aes(x = X.,                                    # <--- Werkt nu met de jaren uit je CSV
           y = Uitstoot,
           fill = Bron)) +
  geom_area(color = "white", linewidth = 0.2) +       # <--- Creëert de witte scheidingslijntjes

  # 5. lijn event analyse 2019 ---
  geom_vline(
  xintercept = 2019,
  color = "darkred",
  linewidth = 1,
  linetype = "dashed"
) +
annotate(
  "text",
  x = 2019,
  y = 285,
  label = "Nitrogen crisis\n2019",
  color = "darkred",
```

```r
  angle = 90,
  vjust = -0.5
) +
  # -----------------------------------------------

  scale_fill_brewer(
  palette = "Set3",
  labels = c(
    "Services, water and construction",
    "Energy sector",
    "Industry and waste",
    "Agriculture, forestry and fisheries",
    "Households",
    "Transport"
  )
) +
  labs(                                               # <--- Namen op de assen en titels
    title = "Nitrogen emissions per source",
    x = "Years",
    y = "Emissions (million kg)",
    fill = "Sources"
  ) +
  theme_bw() +                                        # <--- Grijze rasterlijnen inschakelen
  theme(
    panel.grid.major = element_line(color = "grey80"),
    panel.grid.minor = element_line(color = "grey90"),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```




The graph shows the development of nitrogen emissions from different sources in the Netherlands between 2005 and 2024. Overall, emissions decreased substantially over the observed period, with the largest contribution consistently coming from the transport sector. Emissions from industry, the energy sector, households, agriculture, and services also show a downward trend.

The vertical dashed line indicates the Dutch nitrogen crisis in 2019. After this point, emissions continued to decline across most sectors. This trend coincides with the introduction of stricter environmental measures and, shortly afterwards, the COVID-19 pandemic. However, based on this graph alone, no direct causal relationship can be established between these events and the observed decrease in emissions.

In the graph you see that there is a strong drop in nitrogen emissions across multiple sources from 2019 onward. This can be explained by multiple factors, such as new more strict emissions laws by the Dutch government and also the EU. These laws include a lowering of the speed limit to 100 km/h, causing a sharp drop in vehicle emissions. The laws also include the freezing of a lot of infrastructure and other building projects causing a drop in services and industry emissions. And then from late 2019 and 2020 onward there was the COVID pandemic causing lower emissions in almost all sectors.

# Part 4 - Discussion

## 4.1 Discuss your findings

# Part 5 - Reproducibility

## 5.1 Github repository link

Provide the link to your PUBLIC repository here: https://github.com/OLHHRuyten/Air-pollution-and-health-4.3-NL

## 5.2 Reference list

Use APA referencing throughout your document. For the datasets: AA_BronnenVanFijnstof, BA_BronnenVanStikstof and PolutionIngeneral are from CBS:

Centraal Bureau voor de Statistiek. (2025, 2 september). Uitstoot luchtverontreinigende stoffen lager dan norm voor 2030. https://www.cbs.nl/nl-nl/nieuws/2025/36/uitstoot-luchtverontreinigende-stoffen-lager-dan-norm-voor-2030 Dataset JaargemiddeldeFijnstof Planbureau voor de Leefomgeving. (z.d.). Atlas van de Regio. https://themasites.pbl.nl/atlas-regio/